

Problem-Solution Fit Canvas

Date: 28 June 2026 **Team ID:** **Project Name:** Rising Waters – AI Flood Prediction System

Maximum Marks: 5 Marks

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS]: • People living in flood-prone areas • Farmers • Local residents • Disaster Management Authorities • Government Agencies	2. PROBLEMS / PAINS + ITS FREQUENCY [PR]: • Difficulty predicting floods in advance. • Lack of timely flood warnings. • Property and crop damage. • Threat to human lives. • Floods occur frequently during heavy rainfall.
3. TRIGGERS TO ACT [TR]: • Heavy rainfall alerts. • Rising river water levels. • Government weather warnings. • Previous flood experiences.	4. EMOTIONS BEFORE / AFTER [EM]: Before: • Fear • Anxiety • Confusion After: • Relief after receiving prediction • Better preparedness • Increased confidence
5. AVAILABLE SOLUTIONS PROS & CONS [AS]: Existing Solutions: • Weather forecasting apps • News channels • Government alerts Pros: • Easily accessible • Provide weather information Cons: • Not location-specific • No personalized flood prediction • Limited early risk assessment	6. CUSTOMER LIMITATIONS [CL]: • Limited technical knowledge • Internet connectivity issues • Lack of awareness • Delayed official information
7. BEHAVIOR + ITS INTENSITY [BE]: • Regularly checks weather updates. • Follows local news. • Monitors rainfall. • Seeks early flood information.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: • Mobile Apps • Government Websites • Social Media • News Portals OFFLINE: • Television • Radio • Local Authorities • Community Announcements
9. PROBLEM ROOT / CAUSE [RC]: • Climate change • Heavy rainfall • Poor drainage • Deforestation • River overflow • Urbanization	10. YOUR SOLUTION [SL]: Rising Waters is an AI-powered Flood Prediction System that uses Machine Learning (XGBoost) to analyze environmental parameters and predict flood risk. The system provides early warnings, flood risk percentages, and safety recommendations to help users make informed decisions and improve disaster preparedness.

